ASSESSMENT A2 · STUDENT PACKAGE

AI Systems Essay

Week 6 · Individual assessment

SUBMISSION	WEIGHT	LENGTH
PDF via LMS, before your Week 6 session	15% of Semester 1 grade	500 words · 5 paragraphs

What this assessment is testing

A2 asks you to look at the real world of AI: not theory, not hypothetical scenarios, but named systems and organisations. One went well. One went badly. Your job is to explain why.

It tests three things.

1. Can you identify a root cause, not just a symptom?

When an AI system fails, the visible problem is rarely the actual cause. *'The AI got it wrong'* is a symptom. A root cause is the specific technical or systemic failure that produced the wrong output: biased training data, a deployment without validation, an oversight structure with no mechanism to catch errors. You are expected to explain how the failure happened at a system level.

2. Can you name who was accountable?

Accountability is not the same as blame. It is the question of who had the authority and the responsibility to prevent the failure, and did not. The answer must be a specific role, organisation or institution. *'The AI company'* is not an answer. *'The procurement team at the hospital that adopted the system without independent validation'* is.

3. Can you apply a governance framework, not just cite one?

Naming the EU AI Act is not the same as applying it. Applying it means identifying the principle or category that covers this case and explaining exactly how it would have prevented the failure. A framework mentioned in passing earns nothing; a framework applied to a specific failure through a specific mechanism is what the marks are for.

The task

Write a 500-word essay on one AI success case and one AI failure case. Choose systems you have read about during Weeks 3–5, or ones you have come across in the news. Name the system and the organisation. Your analysis is what is being marked, not your research.

The essay has five paragraphs:

Para	Section	What to cover
1	Introduction	Name both cases and the organisations involved. State in one sentence why you chose them.

Para	Section	What to cover
2	Success case	What the system did, what made it work, and what conditions enabled it.
3	Failure case	What went wrong. The root cause at a technical or systemic level, explained as a mechanism (for example biased training data, a flawed validation pipeline, deployment without audit). Who was specifically accountable.
4	Recommendation and governance	One specific change that would have prevented the failure, framed in terms of the current AI stack (for example RAG, validation layers, output monitoring, human-in-the-loop design). Name a governance framework, identify the relevant principle or category (for example high-risk classification, transparency obligation, bias testing requirement) and explain how it applies to this case.
5	Conclusion	One sentence: what does this tell you about working with AI professionally?

> **A note on frameworks**
> You are not expected to memorise article numbers. What matters is that you can identify which part of a framework applies to your case and explain the connection clearly. The EU AI Act, the NIST AI Risk Management Framework and the US Executive Order on AI are all covered in Weeks 3–5.

How it is marked

Criterion	Marks	What we are looking for
Cases: named system and organisation	15	Named systems, named organisations. No vague or hypothetical references.
Root cause: technical or systemic level	25	Data bias, model design, deployment error or oversight gap, explained as a mechanism. Not ‘the AI got it wrong’.
Accountability: named responsible party	20	A specific role, organisation or institution. Not ‘the AI’ or ‘the company’.
Recommendation: concrete, connected to cause	20	Implementable and directly addresses the root cause. Not ‘test more carefully’.
Governance: framework named and applied	20	Framework named, relevant principle or category explained, applied to this specific case.
Total	100	**Weighted to 15% of the semester grade.**

Common mistakes to avoid

Too vague: will lose marks	Specific enough: this is the standard
An AI system at a hospital gave wrong advice	IBM Watson for Oncology at MD Anderson Cancer Center gave chemotherapy recommendations that contradicted physician judgement
The AI was biased because of the data	The training data consisted of historical arrest records that reflected decades of racially unequal policing, causing the model to associate race with reoffending risk
The company is accountable	Northpointe (the vendor) is accountable for not disclosing the algorithm weights; the Wisconsin court system is accountable for not requiring independent validation before deployment
The EU AI Act would have helped	The EU AI Act classifies judicial AI as high-risk, requiring bias testing before deployment. This would have identified the false-positive disparity before the system was used in sentencing
AI developers should test more carefully	Courts should have required the vendor to provide independent bias audit results before allowing the system to influence sentencing recommendations

Submission template

Write your essay below, then export as PDF

Complete all five paragraphs. Target: 500 words in total. Export this document as a PDF and submit via the LMS before your Week 6 session.

STUDENT NAME	STUDENT ID
BATCH / SECTION	DATE SUBMITTED

Cases	
Success case chosen	
Failure case chosen	

Suggested word count per paragraph

Para 1 · Intro	Para 2 · Success	Para 3 · Failure	Para 4 · Recommendation	Para 5 · Conclusion
~60 words	~120 words	~150 words	~120 words	~50 words (one sentence)

Your essay

Paragraph 1: Introduction

Name both cases and both organisations. State in one sentence why you chose them. Both names must appear before the end of this paragraph.

Paragraph 2: Success case

What the system did, what made it work, and what specific conditions enabled it to succeed.

Paragraph 3: Failure case

What went wrong. The root cause at a technical or systemic level, not 'the AI got it wrong'. Who was specifically accountable, and why.

Paragraph 4: Recommendation and governance

One specific change that would have prevented the failure. Name a governance framework and identify the principle or category that applies to your case.

Paragraph 5: Conclusion

One sentence only. What does this tell you about working with AI professionally? Do not repeat your recommendation.

Pre-submission checklist

- [ ] Both cases name a specific system and a specific organisation, with no hypothetical or vague references.
- [ ] The root cause in paragraph 3 goes deeper than ‘the AI got it wrong’ and names the specific technical or systemic failure.
- [ ] The accountable party in paragraph 3 is a specific role, organisation or institution, not ‘the AI’ or ‘the company’.
- [ ] The recommendation in paragraph 4 is specific and directly connected to the root cause you identified.
- [ ] A governance framework is named in paragraph 4, and a specific principle or category is identified and applied to your case.
- [ ] The conclusion is one sentence and states a professional insight; it does not repeat the recommendation.
- [ ] Total word count is between 400 and 600 words.